

AI-H - 2026Spring - Practice-VIII - Supervised Learning: Movie Profit Prediction Model Comparison

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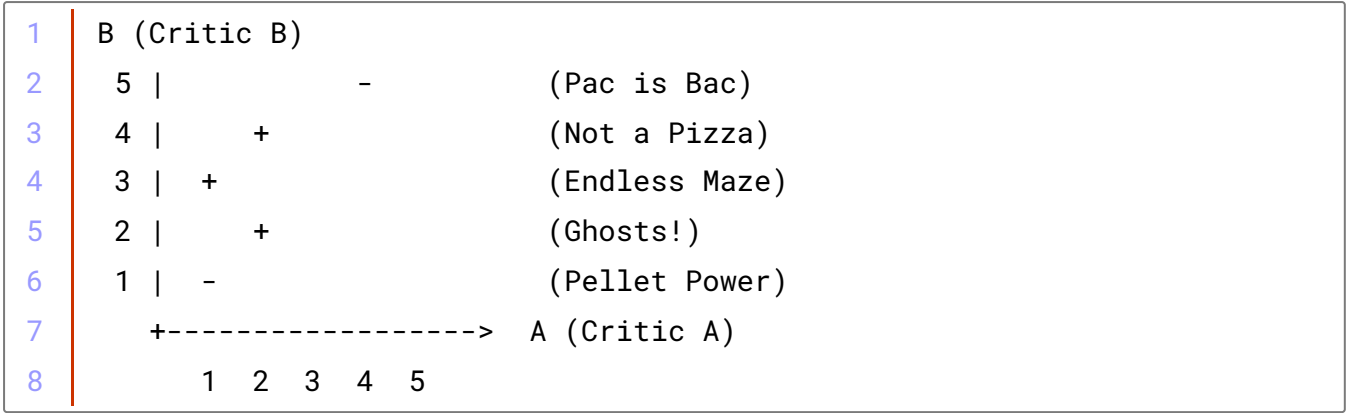
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Dataset Overview

Movie Name	A Score (f_1)	B Score (f_2)	Profit?	Label y
Pellet Power	1	1	No	-1
Ghosts!	3	2	Yes	+1
Pac is Bac	4	5	No	-1
Not a Pizza	3	4	Yes	+1
Endless Maze	2	3	Yes	+1

Task 1: Data Visualization and Linear Separability Analysis

1.1 Scatter Plot (conceptual sketch)



- "+" denotes profitable (Yes): (3, 2), (3, 4), (2, 3)
- "-" denotes non-profitable (No): (1, 1), (4, 5)

1.2 Linear Separability Proof

Observe that the two negative points (1, 1) and (4, 5) lie on the line

$$4x - 3y = 1.$$

Parameterize the segment between them as $(1 + 3t, 1 + 4t)$, $t \in [0, 1]$. Setting $x = 3$ gives $t = 2/3$, yielding the point $(3, 11/3) \approx (3, 3.67)$, which lies **inside** the convex hull of the positive points (the triangle with vertices (3, 2), (3, 4), (2, 3)).

Because the **convex hulls of the two classes intersect**, by the Separating Hyperplane Theorem, **no linear boundary can separate them**. The dataset is **not linearly separable**.

Intuitively, both low scores (1, 1) and high scores (4, 5) indicate failure, while medium scores predict success — this resembles an **XOR-like** pattern.

1.3 Which Model is More Suitable?

Model	Analysis	Verdict
Linear SVM	Cannot handle non-linear separability without kernels. Soft margin helps but cannot achieve perfect accuracy linearly.	X (without kernel)
Neural Network (NN)	Can learn arbitrary non-linear boundaries, but with only 5 samples it is prone to severe overfitting.	△
Decision Tree (DT)	Splits axis-aligned; naturally captures "medium scores → Yes, extreme scores → No" with few splits. Interpretable.	✓

Recommendation: A **Decision Tree** is most suitable given the small dataset size and the non-linear, axis-aligned nature of the decision boundary. An SVM with an RBF kernel is a close second.

Task 2: Perceptron Model Training and Feature Selection

2.1 Feature Representation

Each sample is represented as

$$\mathbf{x}_i = (f_0, f_1, f_2)^\top = (1, A_i, B_i)^\top, \quad y_i \in \{-1, +1\}.$$

2.2 Weight Update Rule

Perceptron prediction: $\hat{y} = \text{sign}(\mathbf{w}^\top \mathbf{x})$.

If $\hat{y}_i \neq y_i$, update:

$$\mathbf{w} \leftarrow \mathbf{w} + y_i \mathbf{x}_i.$$

If correctly classified, leave $\mathbf{w}$ unchanged.

2.3 First Update

Initialize $\mathbf{w}^{(0)} = (0, 0, 0)^\top$.

Take the first training example — **Pellet Power**: $\mathbf{x}_1 = (1, 1, 1)^\top$; $y_1 = -1$.

$$\mathbf{w}^{(0)\top} \mathbf{x}_1 = 0 \cdot 1 + 0 \cdot 1 + 0 \cdot 1 = 0.$$

$\text{sign}(0)$ is conventionally treated as $+1$, which disagrees with $y_1 = -1$. Thus we update:

$$\mathbf{w}^{(1)} = \mathbf{w}^{(0)} + y_1 \mathbf{x}_1 = (0, 0, 0)^\top + (-1)(1, 1, 1)^\top = (-1, -1, -1)^\top.$$

$\mathbf{w}^{(1)} = (-1, -1, -1)^\top$

2.4 Can a Perceptron Perfectly Classify These Rules?

(i) $A + B > 8 \Rightarrow \text{Success}$

Set $w_0 = -8$, $w_1 = 1$, $w_2 = 1$. Then

$$\mathbf{w}^\top \mathbf{x} = -8 + A + B > 0 \iff A + B > 8.$$

This is a single linear half-plane.

✓ **Yes — a perceptron can represent this perfectly.**

(ii) Success iff each critic gives a score of 2 or 3

Positive region = $(A, B) : A \in 2, 3 \text{ and } B \in 2, 3$.

Consider the horizontal line $B = 2$: points $(1, 2)$ $(-)$, $(2, 2)$ $(+)$, $(4, 2)$ $(-)$. A positive point lies strictly between two negatives on the same line — impossible to separate with a linear boundary in the original feature space.

✗ **No — the region is a bounded rectangle, which cannot be expressed as a single linear half-space.** It would require at least the intersection of 4 linear inequalities (i.e., multiple perceptrons or a non-linear model).

(iii) Success iff both critics agree ($A = B$)

The positive set is the line $A = B$ itself (a measure-zero set); everywhere else is negative. A single linear decision boundary $\mathbf{w}^\top \mathbf{x} = 0$ divides the plane into **two** half-spaces, but both "above" and "below" the diagonal contain negatives.

✗ No — a perceptron cannot realize this XOR-like pattern with only the raw features (A, B) .

Remark: Scenarios (ii) and (iii) become perceptron-learnable after a non-linear feature expansion (e.g., polynomial kernels or additional hand-crafted features like $|A - B|$).

Task 3: Soft Margin SVM Analysis

3.1 Formulation

For non-separable data, the **soft margin SVM** introduces slack variables $\xi_i \geq 0$:

$$\min_{\mathbf{w}, b, \xi} \quad \frac{1}{2} |\mathbf{w}|^2 + C \sum_{i=1}^n \xi_i$$

$$\text{s.t.} \quad y_i(\mathbf{w}^\top \mathbf{x}_i + b) \geq 1 - \xi_i, \quad \xi_i \geq 0, \quad \forall i.$$

- $\xi_i = 0$: sample is correctly classified outside the margin.
- $0 < \xi_i \leq 1$: sample is inside the margin but correctly classified.
- $\xi_i > 1$: sample is **misclassified**.

3.2 Role of the Regularization Parameter C

Regime	Effect
Large C (e.g., $C \rightarrow \infty$)	Heavy penalty on $\xi_i \Rightarrow$ few training errors allowed $\Rightarrow$ narrower margin, decision boundary closely fits training data $\Rightarrow$ risk of overfitting .
Small C (e.g., $C \rightarrow 0$)	Low penalty on $\xi_i \Rightarrow$ more violations tolerated $\Rightarrow$ wider margin, smoother boundary $\Rightarrow$ risk of underfitting .

Thus C is the dial that controls the **bias–variance trade-off**.

3.3 Trade-off Between Margin and Error

The objective has two competing terms:

$$\underbrace{\frac{1}{2}|\mathbf{w}|^2}_{\text{*inverse of margin width;}} + C \underbrace{\sum_i \xi_i}_{\text{*total classification error}}$$

- Maximizing the margin ($|\mathbf{w}|^2$ small) improves generalization (VC-theoretically).
- Minimizing training error ($\sum \xi_i$ small) improves fit.

C sets the relative weight. In our 5-point dataset — which is not linearly separable — a **moderate C** (selected via cross-validation) would yield a reasonable margin while permitting the two negative "outliers" (1, 1) and (4, 5) to violate the margin, at the cost of at least one training error. Using a **non-linear kernel** (e.g., RBF) would allow soft-margin SVM to classify all five points correctly.

Task 4: Decision Tree Model Construction

4.1 Splitting Criterion — Information Gain

Total entropy of the root (3 Yes, 2 No):

$$H(S) = -\frac{3}{5}\log_2 \frac{3}{5} - \frac{2}{5}\log_2 \frac{2}{5} \approx 0.971 \text{ bits.}$$

We evaluate candidate splits:

Split	Left subset	Right subset	Weighted Entropy	Info Gain
$A \leq 1$	(1, 1, -)	3Y, 1N	$\frac{1}{5}(0) + \frac{4}{5}(0.811) = 0.649$	0.322
$A \leq 2$	1Y, 1N	2Y, 1N	0.951	0.020
$A \leq 3$	3Y, 1N	(4, 5, -)	0.649	0.322
$B \leq 1$	(1, 1, -)	3Y, 1N	0.649	0.322
$B \leq 3$	2Y, 1N	1Y, 1N	0.951	0.020
$B \leq 4$	3Y, 1N	(4, 5, -)	0.649	0.322

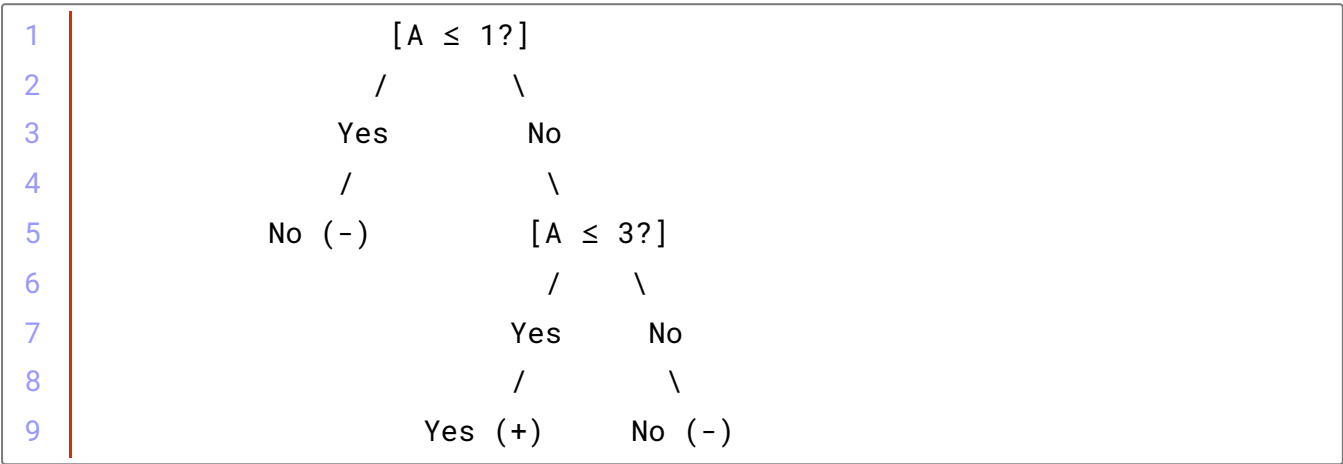
Tied best splits with gain 0.322. Choose $A \leq 1$ (others are equivalent).

4.2 Tree Construction

Root: split on $A \leq 1$.

- Left** ($A \leq 1$): one sample, Pellet Power $\Rightarrow$ **No** ✓
- Right** ($A > 1$): (3, 2, +), (4, 5, -), (3, 4, +), (2, 3, +), entropy = 0.811.
 - Split on $A \leq 3$: Left = all 3 positives (pure), Right = (4, 5, -) (pure). Info gain = 0.811.

Final tree:



In words:

$\hat{y} = \{\text{No}, \quad A = 1 \text{ Yes}, \quad 2 \leq A \leq 3 \text{ No}, \quad A \geq 4$

This depth-2 tree achieves **100% training accuracy**.

4.3 Depth and Pruning

- **Depth too shallow** $\Rightarrow$ underfits (cannot capture medium-vs-extreme pattern).
- **Depth too deep** $\Rightarrow$ each leaf holds very few samples (with only 5 points, a depth-3 tree would trivially memorize noise) $\Rightarrow$ **overfitting**.
- **Pruning** combats overfitting:
 - **Pre-pruning**: stop splitting when info gain $<$ threshold, leaf size $< n_{\min}$, or max depth reached.
 - **Post-pruning** (e.g., cost-complexity pruning): grow a full tree, then collapse subtrees whose removal does not significantly increase validation error, minimizing $R_{\alpha}(T) = R(T) + \alpha \cdot |T|_{\text{leaves}}$.

For our tiny dataset, a depth limit of 2 (as derived) is already an implicit form of pruning and is appropriate.

Task 5: Model Comparison and Selection

5.1 Side-by-Side Comparison

Aspect	SVM (soft margin / RBF)	Neural Network	Decision Tree
Handles non-linearity	✓ via kernel	✓ via hidden layers	✓ axis-aligned splits
Performance on 5 samples	Reasonable, but kernel tuning needed	Poor — severe overfitting with so few points	Excellent — splits are natural
Interpretability	Low (especially with kernels)	Very low (black box)	Very high (if-else rules)
Training cost	Moderate ($O(n^2)$ – $O(n^3)$)	High; many hyperparameters	Low
Sensitivity to hyperparams	High (C , kernel, γ)	Very high (architecture, lr, ...)	Low (depth, min-samples)
Robustness to outliers	Soft margin tolerates them	Can memorize them	Isolates them in leaves
Data requirement	Small–medium	Large	Small

5.2 Comprehensive Evaluation

For this **specific problem** (5 samples, 2 features, non-linear XOR-like structure, need for interpretability to justify greenlighting a film):

- A **Neural Network** is inappropriate — only 5 training examples cannot support reliable weight estimation; it would either underfit drastically or memorize the set.
- A **Kernel SVM** (RBF) would work in practice, but requires tuning C and γ via cross-validation, which is statistically unstable with 5 points, and the model is essentially a black box to a film executive.

- A **Decision Tree** is the clear winner: it achieves perfect training accuracy with a 2-level depth, produces **human-readable rules** ("the film succeeds iff at least one critic gives a middling score of 2 or 3"), and naturally captures the non-linear pattern without any kernel trick.

5.3 Final Recommendation

Choose the Decision Tree.

Reasons:

1. **Accuracy:** perfectly fits the non-linearly-separable training data with minimal depth.
2. **Interpretability:** decision rules can be communicated directly to producers and writers.
3. **Data efficiency:** works well even in small-sample regimes where NN/SVM may be unstable.
4. **Low computational cost** and **few hyperparameters**, making it easy to tune and deploy.

If the dataset were expanded to hundreds or thousands of movies, an **ensemble method** such as **Random Forest** or **Gradient Boosted Trees** would be the preferred upgrade, combining the interpretability of trees with lower variance. A kernel SVM or a small NN with regularization would become competitive alternatives in that regime.